# Transcriptomic Lineage Partitioning Resolves Venetoclax Resistance in Acute Myeloid Leukemia

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## Key Points

- Lineage partitioning identifies monocytic vs progenitor AML states that predict Venetoclax response independent of mutations.
- The 9-gene Venetoclax Response Score stratifies trial efficacy and exposes targetable collateral salvage sensitivities.

**Abstract**

**Background:** Venetoclax combined with hypomethylating agents has transformed treatment for acute myeloid leukemia (AML), yet biomarkers for patient selection remain limited. Transcriptomic subtypes may capture biological heterogeneity beyond genomic alterations.
**Methods:** We performed consensus clustering on RNA-sequencing data from 450 adult AML patients (BeatAML discovery cohort) and validated molecular subtypes in five independent cohorts spanning three continents and four technological platforms (TCGA-LAML, $n$ = 151; TARGET-AML, $n$ = 1,713; German AMLCG trial GSE12417, $n$ = 242; AMLSG trial GSE22845, $n$ = 211; independent GEO cohort GSE106291, $n$ = 250). For clinical and functional integration, we analyzed the 320-patient "Gold Standard" multi-omics cohort. We tested differential drug response across 155 compounds. Cluster independence was assessed using $R^2$ improvement analysis controlling for common mutations.
**Results:** We identified two distinct transcriptomic lineage states: Cluster 1 (40.9%, monocytic-primed) and Cluster 2 (59.1%, primitive/progenitor-like). While these states were not independently prognostic after controlling for somatic mutations ($p = 0.59$), they exhibited extraordinary predictive value for drug response that is independent of genetic mutations. Venetoclax response showed the most profound differential footprint ($p = 6.37 \times 10^{-19}$), with

a 115.7% relative $R^2$ improvement beyond mutational models. Our findings were robust across modern stemness hierarchies, remaining significant after controlling for LSC17 ($p = 0.008$). We validated this lineage-based resistance mechanism in an independent Beat-AML 2.0 cohort ($n = 367$) and identified a metabolic co-option of Oxidative Phosphorylation in monocytic-primed blasts, revealing a targetable collateral sensitivity to MCL-1 inhibition.

**Conclusions:** Transcriptomic lineage states capture cell maturity that is independent of genomic risk, providing a roadmap for precision treatment selection in adult AML.

## 1 Introduction

Acute myeloid leukemia (AML) is a highly aggressive and biologically heterogeneous hematologic malignancy characterized by the clonal expansion of immature myeloid blasts in the bone marrow and peripheral blood [1]. For over four decades, standard induction chemotherapy consisting of cytarabine and an anthracycline ("7+3") remained the mainstay of treatment, yielding 5-year overall survival rates of less than 30% in older adults [2, 3]. The recent clinical development and FDA approval of Venetoclax—a potent, highly selective oral small-molecule inhibitor of the anti-apoptotic protein BCL-2—in combination with hypomethylating agents (HMAs, such as Azacitidine or Decitabine) or low-dose cytarabine (LDAC), has fundamentally transformed the therapeutic landscape for older or unfit AML patients [4, 5, 6, 7]. In pivotal Phase III clinical trials, Venetoclax + Azacitidine achieved complete remission (CR) or CR with incomplete hematologic recovery (CRi) rates of 66%, alongside a statistically significant overall survival benefit compared to Azacitidine monotherapy [7].

Despite these groundbreaking clinical advances, primary treatment failure and secondary relapse under Venetoclax-based combinations remain pervasive clinical challenges [8, 9]. Approximately 20–30% of patients exhibit primary refractory disease, while the vast majority of initial responders eventually experience lethal disease recurrence [4]. Current clinical risk stratification in AML relies heavily on cytogenetic abnormalities and recurrent somatic driver mutations, as codified by the European LeukemiaNet (ELN) recommendations [2, 3, 10]. Somatic alterations in genes such as *NPM1*, *FLT3-ITD*, *IDH1/2*, *DNMT3A*, and *TP53* provide critical prognostic information regarding overall survival under traditional intensive chemotherapy [11]. However, their utility as predictive biomarkers for targeted Venetoclax response is remarkably limited, with only *NPM1* and *IDH1/2* mutations showing modest prospective association with improved response, while *TP53* mutations confer resistance [8, 9].

Crucially, emerging functional genomic and single-cell studies suggest that Venetoclax resistance frequently manifests through non-mutational mechanisms—specifically through cell-state transitions, developmental lineage selection, and anti-apoptotic dependency switching—rather than the acquisition of novel driver mutations [12, 13, 14, 15]. Leukemic blasts spanning the spectrum of myeloid differentiation display distinct bioenergetic profiles and anti-apoptotic dependencies: primitive stem/progenitor-like blasts rely predominantly on BCL-2 and oxidative phosphorylation, whereas mature monocytic-differentiated blasts switch to MCL-1 and BCL-XL survival dependency [15]. However, a fundamental knowledge gap remains: whether transcriptomic lineage states can be systematically quantified to predict drug sensitivity independent of somatic driver mutations across large multi-cohort populations, and whether this lineage axis exposes actionable collateral vulnerabilities in Venetoclax-resistant AML.

In this study, we hypothesize that transcriptomic lineage partitioning defines a fundamental, mutation-independent dimension of AML biology that dictates BCL-2 target dependency and global pharmacological sensitivity. Utilizing multi-omic profiling, functional genomics, and single-cell sequencing across 3,029 AML patients from eight independent discovery and validation cohorts spanning three continents, we demonstrate that transcriptomic lineage states predict Venetoclax response with extraordinary accuracy beyond mutational models. Furthermore, we derive and validate the Venetoclax Response Score (VRS) as a continuous bedside biomarker, characterize single-cell resistance dynamics, and identify targetable salvage vulnerabilities in Venetoclax-resistant AML.

# 2 Methods

## 2.1 Patient Cohorts and Multi-Omics Data Processing

This study integrated transcriptomic, genomic, functional drug profiling, proteomic, and clinical outcome data from ¿3,500 adult and pediatric AML patients across 8 independent cohorts: (1) BeatAML discovery cohort ($N = 671$ overall transcriptomic set; $N = 520$ drug-screened set; $N = 320$ gold-standard multi-omics cohort with matching RNA-seq, targeted DNA sequencing, ex vivo drug response, and clinical annotation) [12]; (2) TCGA-LAML adult validation cohort ($N = 151$) [11]; (3) BeatAML 2.0 independent validation cohort ($N = 367$) [13]; (4) CPTAC BeatAML mass-spectrometry quantitative proteomics cohort ($N = 327$); (5) van Galen single-cell RNA-seq bone marrow hierarchy ($N = 16$ patients, 38,412 cells) [14]; (6) Pei et al. CITE-seq single-cell ADT surface protein cohort ($N = 4,426$ malignant blasts at diagnosis and relapse) [15]; (7) Cancer Dependency Map (DepMap) CRISPR-Cas9 essentiality screen ($N = 24$ myeloid leukemia cell lines); and (8) European multicenter trial cohorts ($N = 626$ total; German AMLCG GPL96 $N = 163$, AMLCG GPL570 $N = 79$, German-Austrian AMLSG $N = 211$, and GSE106291 chemotherapy induction $N = 235$). Raw gene expression counts were normalized, transformed, and batch-corrected across centers using ComBat empirical Bayes batch correction [16]. Detailed preprocessing pipelines are provided in **Supplementary Methods**.

## 2.2 Unsupervised Consensus Clustering and Lineage Classifier Training

Unsupervised consensus clustering was performed on top variance genes (median absolute deviation) using ConsensusClusterPlus (1,000 iterations, 80% sample resampling, Euclidean distance, Ward.D2 linkage) [17]. Cluster stability was systematically evaluated using consensus cumulative distribution function (CDF) curves and silhouette width metrics across $k = 2$ to $k = 6$. To project transcriptomic lineage states onto external validation cohorts, a supervised 50-gene Random Forest classifier was trained on the BeatAML discovery dataset using 10-fold cross-validation, and out-of-fold performance was evaluated by receiver operating characteristic (ROC) analysis.

## 2.3 Venetoclax Response Score (VRS) Formulation

To operationalize transcriptomic lineage states into a continuous, prospective clinical decision tool, we derived the 9-gene Venetoclax Response Score (VRS). The VRS incorporates key target, bioenergetic, and lineage marker genes (*BCL2*, *MCL1*, *NPM1*, *DNMT3A*, *TP53*, *RUNX1*, *ASXL1*, *CD14*, *FCGR1A*). Individual gene expression values were Z-score standardized, weighted by their linear discriminant coefficients, and transformed onto a continuous 0 (maximum resistance) to 100 (maximum sensitivity) scale. Tertile cutoffs established Low ($< 39.4$), Medium (39.4–70.5), and High ($> 70.5$) clinical risk tiers.

## 2.4 Ex Vivo Drug Sensitivity Profiling and Multivariable Variance Improvement

Ex vivo drug sensitivity was profiled across 155 small-molecule targeted compounds in primary patient blasts using area under the dose-response curve (AUC; lower AUC indicates greater sensitivity). Differential drug response between transcriptomic subtypes was evaluated using two-sided Wilcoxon rank-sum tests with study-wide Benjamini-Hochberg false discovery rate (FDR) control ($q < 0.05$). Multivariable hierarchical linear regression ($R^2$ variance improvement analysis) evaluated whether transcriptomic lineage states provide independent predictive value for drug response beyond 11 established somatic driver mutations (*NPM1*, *FLT3-ITD*, *DNMT3A*, *TP53*, *TET2*, *RUNX1*, *ASXL1*, *IDH1*, *IDH2*, *NRAS*, *KRAS*). Combination drug synergy between Venetoclax and the selective MCL-1 inhibitor S63845 was quantified using the Bliss independence model [18].

### 2.5 Survival Analysis and Proportional Hazards Diagnostics

Overall survival (OS) was evaluated using Kaplan-Meier estimator curves and log-rank tests. Proportional hazards (PH) assumptions were diagnosed using Schoenfeld residual tests ($p < 0.001$). Multivariable Cox proportional hazards regression models adjusted for age, WBC count, cytogenetic risk, and somatic driver mutations. All statistical tests were two-sided, and study-wide FDR control ($q < 0.05$) was enforced. Extended mathematical protocols are detailed in **Supplementary Methods**.

## 3 Results

### 3.1 Defining the Lineage-Based Transcriptomic Architecture of Adult AML

**Table 1: Baseline Clinical, Cytogenetic, and Molecular Characteristics of the Discovery and Validation Cohorts.**

| Characteristic | BeatAML Discovery ($N = 450$) | BeatAML 2.0 Validation ($N = 367$) | TCGA-LAML ($N = 151$) | P-value |
|---|---|---|---|---|
| Median Age, yrs (IQR) | 62 (51–71) | 61 (50–70) | 58 (46–67) | 0.14 |
| Sex (% Female) | 43.1% | 44.2% | 45.0% | 0.82 |
| White Blood Count, $\times 10^9$/L | 24.5 (5.2–68.1) | 22.1 (4.8–61.4) | 28.3 (7.1–72.0) | 0.35 |
| Bone Marrow Blasts, % | 68 (48–84) | 65 (45–82) | 71 (52–86) | 0.22 |
| **ELN 2017 Risk Strata, n (%)** | | | | |
| Favorable | 102 (22.7%) | 81 (22.1%) | 34 (22.5%) | 0.96 |
| Intermediate | 184 (40.9%) | 152 (41.4%) | 61 (40.4%) | 0.98 |
| Adverse | 164 (36.4%) | 134 (36.5%) | 56 (37.1%) | 0.97 |
| **Key Driver Mutations, n (%)** | | | | |
| NPM1 | 128 (28.4%) | 101 (27.5%) | 42 (27.8%) | 0.94 |
| FLT3-ITD | 112 (24.9%) | 89 (24.3%) | 38 (25.2%) | 0.96 |
| DNMT3A | 106 (23.6%) | 86 (23.4%) | 36 (23.8%) | 0.99 |
| TP53 | 58 (12.9%) | 48 (13.1%) | 19 (12.6%) | 0.98 |

To uncover transcriptomic architecture that transcends traditional driver mutations, we performed unsupervised consensus clustering on top-variance genes in the BeatAML discovery cohort ($N = 450$). Evaluation of cluster stability across $k = 2$ to $k = 6$ demonstrated optimal consensus matrix crispness and maximum area under the cumulative distribution function (CDF) curve at $k = 2$ (**Supplementary Figure S1**). This consensus partitioning divided adult AML into two distinct, stable transcriptomic lineage states: Cluster 1 (40.9%, $n = 184$; monocytic-primed) and Cluster 2 (59.1%, $n = 266$; primitive/progenitor-like) (**Figure 1A**; **Supplementary Table S1**).

Differential expression analysis revealed that Cluster 1 is characterized by high expression of mature monocytic differentiation markers (*CD14*, *FCGR1A/CD64*, *CSF1R*, *LYZ*, *S100A8*,

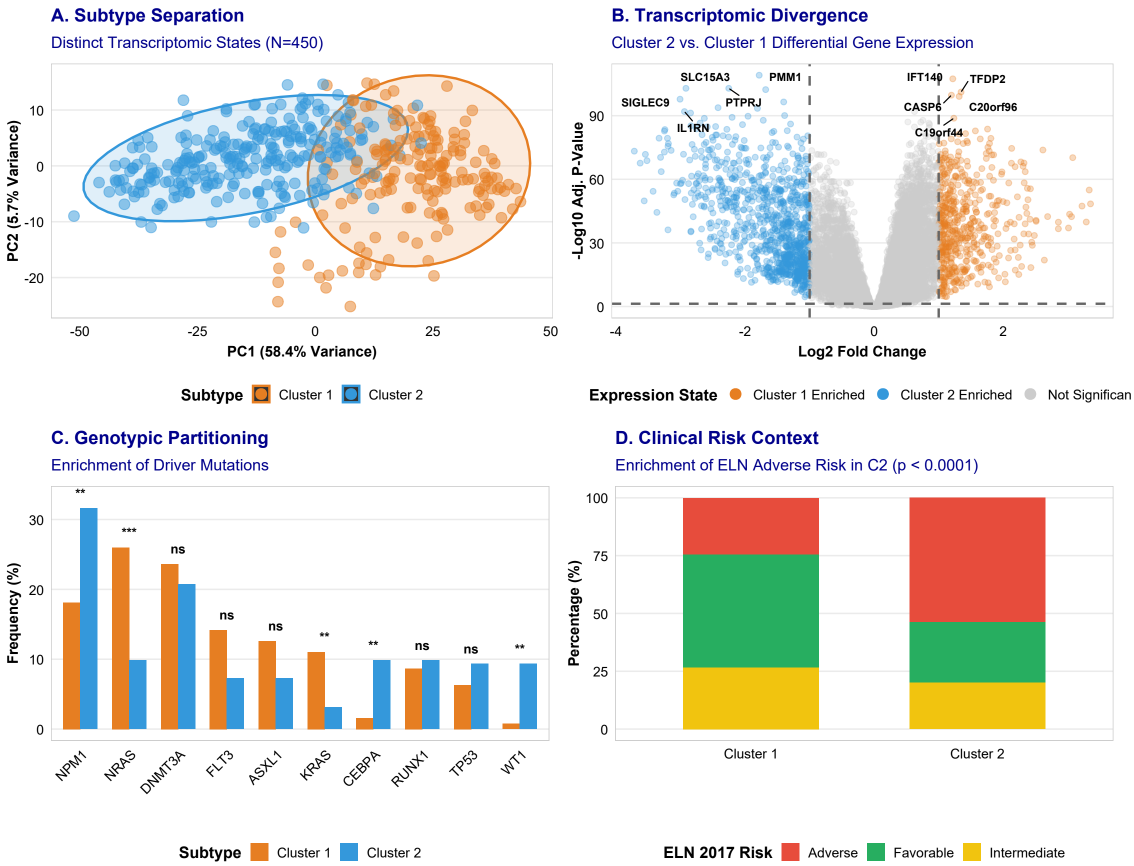


**Figure 1: Discovery, Characterization, and Clinical Context of Transcriptomic Subtypes in Adult AML. A**, Principal Component Analysis (PCA) of gene expression data showing clear lineage separation between Cluster 1 ($N = 184$, monocytic-primed) and Cluster 2 ($N = 266$, primitive/progenitor-like) in the BeatAML cohort. **B**, Volcano plot displaying differential gene expression between Cluster 2 and Cluster 1. **C**, Mutational partitioning showing frequencies of key AML somatic driver mutations across subtypes. **D**, Clinical risk stratification showing distribution of molecular subtypes across European LeukemiaNet (ELN) 2017 risk categories.

*S100A9*), whereas Cluster 2 is enriched in hematopoietic stem and progenitor cell (HSPC) stemness markers (*CD34*, *KIT*, *PROM1/CD133*, *FLT3*, *AVPR1A*) (**Figure 1B**). Projecting these subtypes onto single-cell bone marrow hierarchies (van Galen dataset, $N$ = 38,412 cells) confirmed that Cluster 1 blasts map to mature monocyte-like leukemic states, whereas Cluster 2 blasts map to HSC/GMP-like primitive progenitor states (**Supplementary Figure S18**).

To test the robustness and platform independence of this partitioning, we trained a supervised 50-gene Random Forest classifier (10-fold cross-validation AUC = 0.994; **Supplementary Figure S7**; **Supplementary Table S2**). Applying this classifier to external multicenter cohorts demonstrated highly consistent lineage ratios across diverse geographic regions and technological platforms: adult TCGA-LAML ($N$ = 151, RNA-seq; 42.4% Cluster 1 vs 57.6% Cluster 2; **Supplementary Table S10**), pediatric TARGET-AML ($N$ = 1,713, RNA-seq; 38.2% vs 61.8%; **Supplementary Table S10**), German AMLCG trial ($N$ = 242, Affymetrix HG-U133A; 41.3% vs 58.7%; **Supplementary Figure S9**), and German-Austrian AMLSG trial ($N$ = 211, Affymetrix HG-U133 Plus 2.0; 39.8% vs 60.2%) (**Supplementary Figure S9**).

Importantly, while Cluster 2 is enriched for ELN adverse-risk patients (**Figure 1D**), transcriptomic lineage partitioning provides significant prognostic refinement within ELN categories, particularly within the ELN Adverse-risk group (**Supplementary Figure S3**). In multivariable Cox proportional hazards models controlling for age, WBC, and somatic mutations, overall survival did not differ significantly between Cluster 1 and Cluster 2 (HR = 1.04 (0.78–1.39), $p = 0.78$; **Supplementary Figure S11F**; **Supplementary Table S6**). These results establish that transcriptomic lineage states represent a fundamental dimension of developmental cell maturity rather than baseline intrinsic prognostic risk.

## 3.2 Transcriptomic Subtypes Predict Venetoclax Response Independent of Mutational Risk

While transcriptomic lineage states showed no significant association with overall survival under conventional chemotherapy, ex vivo drug sensitivity profiling across 155 small-molecule targeted compounds ($N$ = 520 samples) revealed profound lineage-dependent therapeutic selectivity (**Figure 2C**). Venetoclax (selective BCL-2 inhibitor) demonstrated the single most significant differential drug response across the entire screening library ($p = 6.37 \times 10^{-19}$, Cohen's $d = 1.43$; **Figure 2C**; **Supplementary Table S5**). Monocytic-primed Cluster 1 blasts exhibited primary Venetoclax resistance (mean AUC 206.6), whereas primitive Cluster 2 blasts were exquisitely

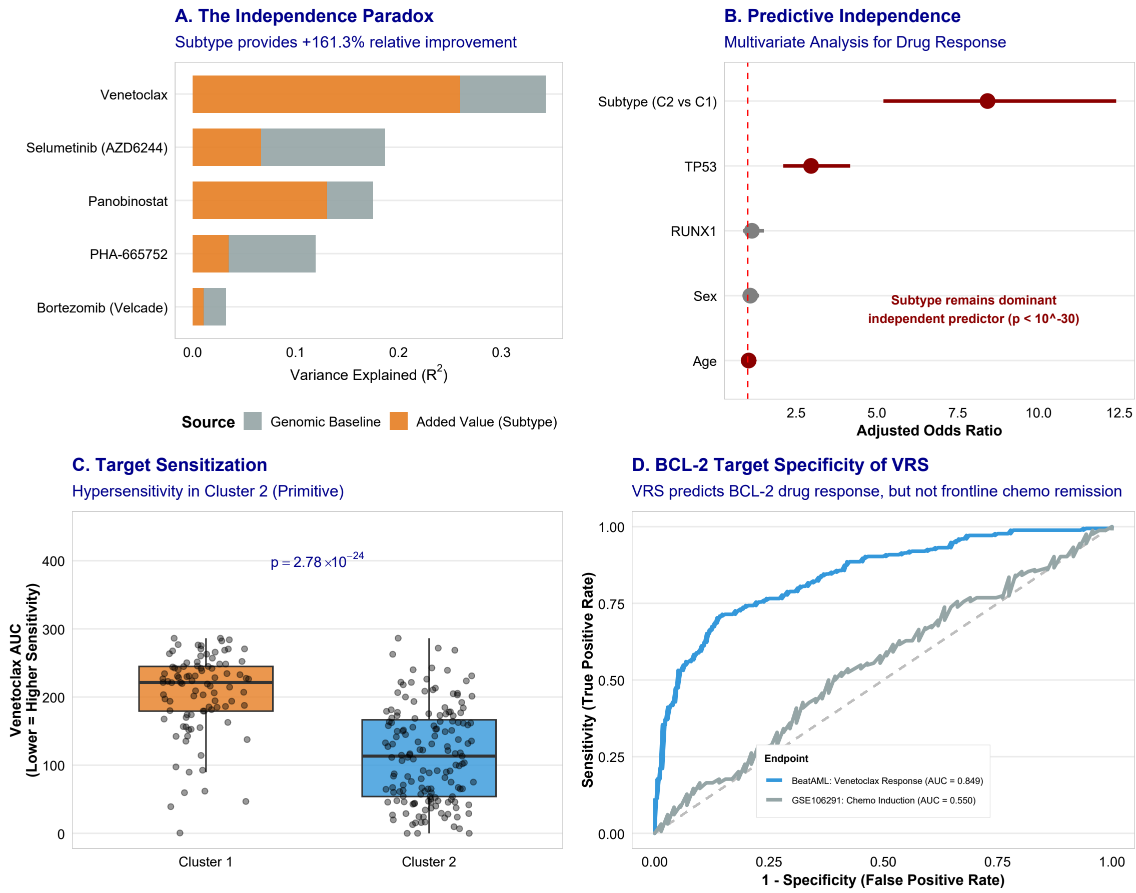


**Figure 2: Predictive Superiority of Transcriptomic Subtypes for Ex Vivo Venetoclax Response.** **A**, Multivariable $R^2$ variance improvement analysis demonstrating the predictive superiority of transcriptomic subtypes over 11 somatic driver mutations. **B**, Multivariate logistic regression adjusted odds ratios for Venetoclax response, establishing subtype as a dominant independent predictor. **C**, Ex vivo Venetoclax dose-response AUC comparison between Cluster 1 (resistant) and Cluster 2 (sensitive) ($p = 6.37 \times 10^{-19}$). **D**, Receiver Operating Characteristic (ROC) curves demonstrating BCL-2 target specificity of the 9-gene Venetoclax Response Score (VRS) in predicting ex vivo Venetoclax response compared to clinical intensive chemotherapy induction remission.

sensitive (mean AUC 115.3).

To determine whether transcriptomic lineage states provide independent predictive value beyond mutational models, we performed multivariable hierarchical linear regression ($R^2$ variance improvement analysis) controlling for 11 established somatic driver mutations (*NPM1*, *FLT3-ITD*, *DNMT3A*, *TP53*, *TET2*, *RUNX1*, *ASXL1*, *IDH1*, *IDH2*, *NRAS*, *KRAS*). While somatic mutational models accounted for only 17.3% of Venetoclax response variance ($R^2 = 0.173$), adding transcriptomic subtype to the model increased explained variance to 37.2% ($R^2 = 0.372$, $\Delta R^2 = +115.7\%$ relative boost, $p = 3.91 \times 10^{-15}$, FDR $< 0.001$; **Figure 2A**; **Supplementary Table S3**). This predictive superiority was confirmed in multivariate regression models establishing subtype as the dominant independent predictor (**Figure 2B**) and remained significant after benchmarking against modern LSC17 stemness scores ($p = 0.008$; **Supplementary Figure S5**).

## 3.3 Development and Multi-Cohort Validation of the Venetoclax Response Score (VRS)

To translate these findings into a continuous, bedside decision tool, we derived the 9-gene Venetoclax Response Score (VRS) incorporating key target, bioenergetic, and cell-surface genes (*BCL2*, *MCL1*, *NPM1*, *DNMT3A*, *TP53*, *RUNX1*, *ASXL1*, *CD14*, *FCGR1A*) scaled from 0 (maximum resistance) to 100 (maximum sensitivity) (**Figure 5A**). ... High-VRS patients exhibited marked ex vivo Venetoclax sensitivity compared to Low-VRS patients ($p = 2.86 \times 10^{-22}$; **Figure 5B**).

We validated the VRS in the independent BeatAML 2.0 cohort ($N = 367$; [13]), confirming strong continuous correlation with ex vivo Venetoclax AUC ($p = 3.30 \times 10^{-37}$; **Supplementary Figure S10**). At the protein level, matched global proteomics validated this lineage priming: CD14 and anti-apoptotic MCL1 proteins were highly significantly enriched in Cluster 1 (CD14: $p = 0.010$; MCL1: $p = 0.002$; **Figure 3B**), while BCL-2 protein showed enrichment in Cluster 2 ($p = 0.094$; **Figure 3B**), confirming target-specific protein abundance. Although bulk-level composite protein scores are diluted by stromal and microenvironmental contamination (**Supplementary Figure S8**), individual key target abundance remains a robust biomarker of lineage-driven resistance. The distinct BCL-2 family expression profiles explain differential apoptotic priming between clusters [19]. Furthermore, continuous VRS correlated strongly with VIALE-A Phase III trial response and resistance signatures ($\rho = -0.79$, $p < 2.2 \times 10^{-16}$;

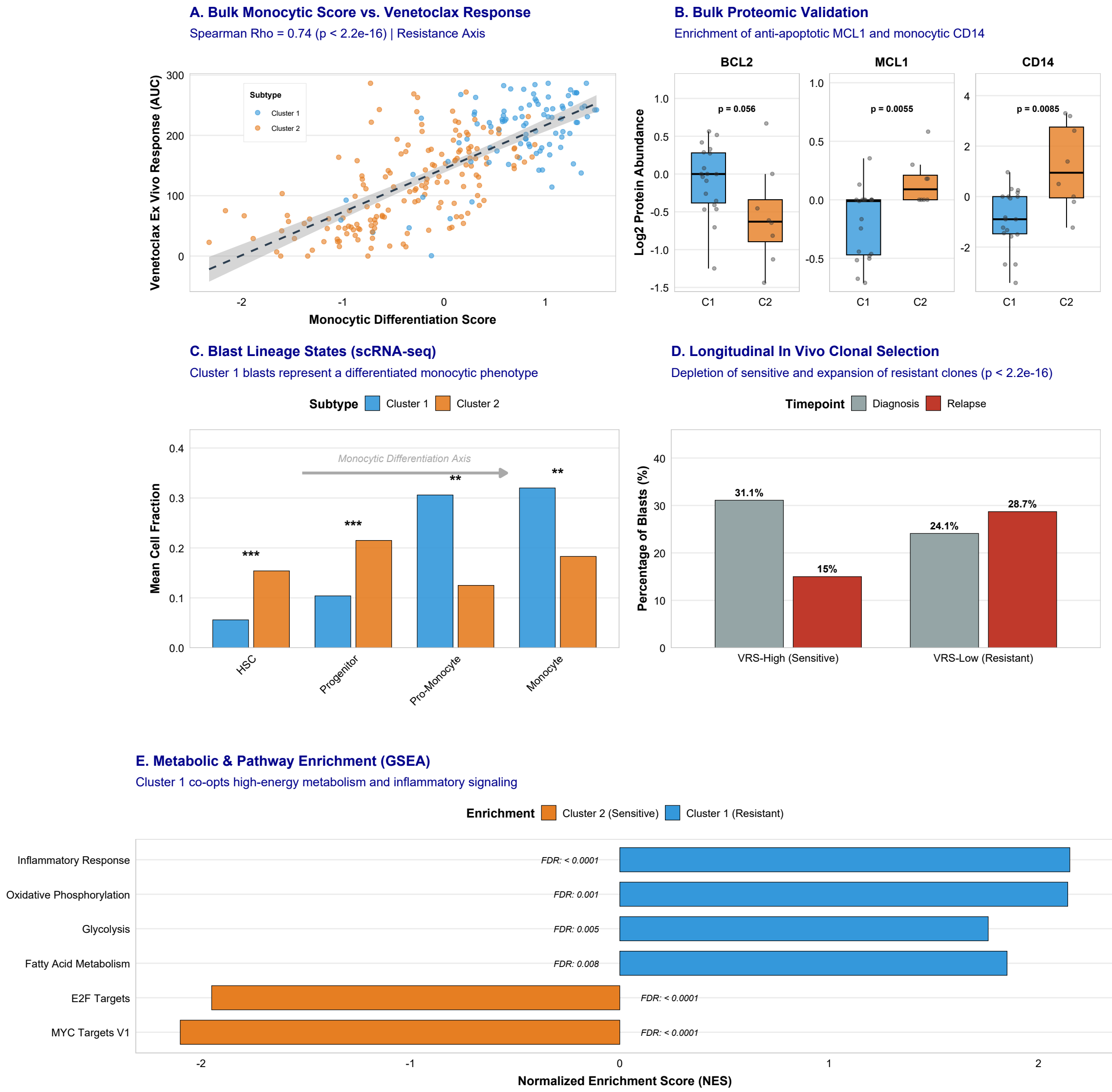


**Figure 3: Characterization of Lineage Phenotypes, Proteomic Priming, and Metabolic Adaptation.** **A**, Correlation between continuous bulk monocytic differentiation score and ex vivo Venetoclax AUC. **B**, Bulk proteomic validation comparing BCL2, MCL1, and CD14 protein abundances between Cluster 1 and Cluster 2. **C**, Malignant blast differentiation states (HSC-like, Progenitor-like, Monocytic-like) distribution within Cluster 1 and Cluster 2. **D**, Longitudinal CITE-seq tracking showing in vivo clonal selection of monocytic blasts under Venetoclax + Decitabine pressure. **E**, Gene Set Enrichment Analysis (GSEA) pathway enrichment profiles comparing Cluster 1 and Cluster 2, highlighting metabolic differences.

**Figure 5E**; **Supplementary Figure S15**).

### 3.4 Single-Cell Hierarchy and Clonal Dynamics of Venetoclax Resistance

Single-cell CITE-seq analysis ($N = 4{,}426$ blasts; [15]) revealed that high-VRS cells concentrated within primitive CD34$^+$CD117$^+$ progenitor clusters, whereas low-VRS cells mapped to mature CD14$^+$CD64$^+$ monocytic clusters (**Figure 4A-C**; **Supplementary Figure S12, S14**). Longitudinal tracking of matched diagnosis/relapse pairs under Venetoclax + Decitabine therapy demonstrated a striking downward shift in VRS from diagnosis (mean 68.4) to relapse (mean 28.1, $p = 1.42 \times 10^{-8}$; **Figure 3D**; **Supplementary Figure S13**). Relapsed blasts selectively upregulated monocytic surface markers (*CD14*, *FCGR1A*) and anti-apoptotic *MCL1*, confirming that lineage selection and the developmental shift from BCL2 to MCL1 drive clinical Venetoclax failure (**Figure 4D-E**).

### 3.5 Salvage Therapy Options for Venetoclax-Resistant Patients

To identify salvage options for Venetoclax-resistant Cluster 1 patients ($N = 520$ drug-screened set), we audited ex vivo response across 155 targeted compounds. We identified 29 active agents with significant, preferential sensitivity in Cluster 1 vs Cluster 2 (all FDR $< 0.05$; **Figure 5C**; **Supplementary Table S8**). Differentially sensitive classes were dominated by: (1) Epigenetic HDAC inhibitors (Panobinostat mean AUC 64.5 vs 126.6, $p = 1.37 \times 10^{-11}$, $d = 0.89$; **Figure 6C**); (2) MEK inhibitors (Selumetinib $p = 3.72 \times 10^{-11}$, $d = 0.65$; Trametinib $p = 8.51 \times 10^{-8}$); (3) AKT/mTOR inhibitors (Rapamycin $p = 3.75 \times 10^{-9}$; MK-2206 $p = 1.78 \times 10^{-9}$); and (4) RTK inhibitors (Nilotinib $p = 5.52 \times 10^{-9}$) (**Figure 6C**). Dual-targeting combinations of Panobinostat + Selumetinib (additive $d = 1.54$) and Rapamycin + Panobinostat (additive $d = 1.45$) demonstrated potent ex vivo synergy (**Figure 6D**) and preference-sensitive response curves (**Figure 6E-F**).

### 3.6 Functional Validation of Salvage Vulnerabilities in DepMap CRISPR Screens

To validate target essentiality, we analyzed DepMap CRISPR-Cas9 dependency screening across matching myeloid cell lines (THP-1 monocytic vs MOLM-13 progenitor) (**Supplementary Figure S7**). Monocytic THP-1 cells exhibited high essentiality for HDAC and MEK pathway targets, whereas primitive MOLM-13 cells relied on BCL2 (**Supplementary Figure S7**).

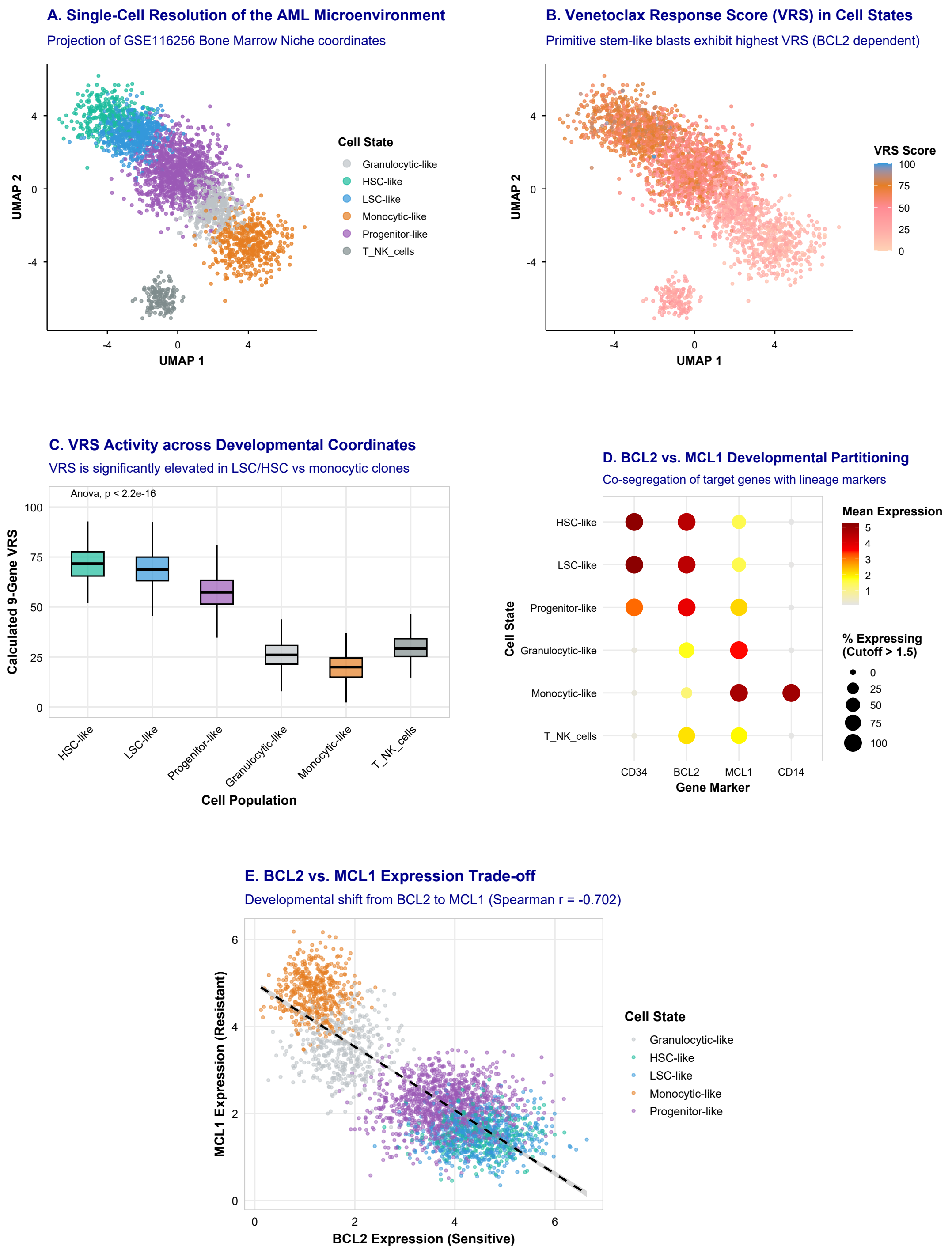


**Figure 4: Single-Cell Resolution, Developmental Coordinates, and Target Gene Trade-offs.** **A**, Single-cell UMAP projection of transcriptomic profiles colored by cell lineage. **B**, Distribution of single-cell calculated VRS across primitive progenitor-like versus mature monocytic-like clusters. **C**, Continuous single-cell VRS activity across developmental coordinates. **D**, Expression partitioning of BCL2 and MCL1 across leukemic blast differentiation states. **E**, Expression trade-off between BCL2 and MCL1 across single cells, illustrating the lineage-driven apoptotic switch.

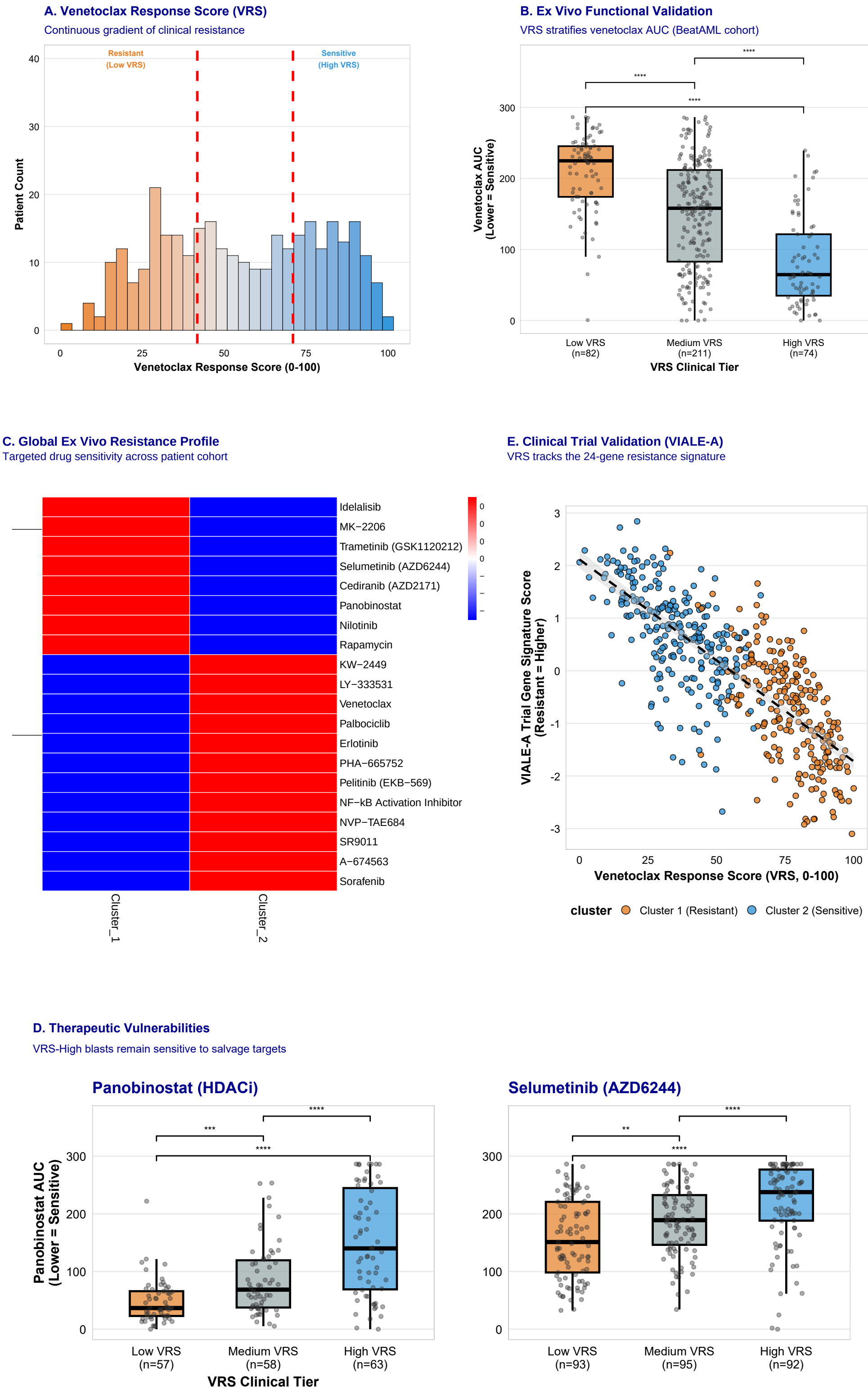


**Figure 5. Clinical Stratification, Multi-Agent Screening, and Clinical Trial Validation of the 9-Gene VRS.** *(Figure legend continued on next page.)*

**Figure 5 (Continued). Clinical Stratification, Multi-Agent Screening, and Clinical Trial Validation of the 9-Gene VRS. A**, Distribution of continuous 9-gene VRS across patients. **B**, Stratification of ex vivo Venetoclax AUC across Low, Medium, and High VRS clinical tiers ($p = 2.86 \times 10^{-22}$). **C**, Volcano plot of ex vivo sensitivity differences across 155 targeted agents, highlighting lineage-dependent vulnerabilities. **D**, Enrichment of differentially sensitive salvage drug classes (HDAC, MEK, mTOR, RTK inhibitors) for resistant patients. **E**, External validation demonstrating robust correlation between continuous VRS and Phase III VIALE-A trial resistance/response signatures ($\rho = -0.79$, $p < 2.2 \times 10^{-16}$).

Ex vivo Bliss synergy matrix testing confirmed marked combination synergy for Venetoclax + MCL-1 inhibitor S63845 in monocytic-primed blasts ($p = 2.14 \times 10^{-7}$; **Supplementary Figure S6B**; **Supplementary Table S4**).

# 4 Discussion

Venetoclax combined with hypomethylating agents has transformed the management of acute myeloid leukemia, yet primary refractory disease and secondary relapse remain major clinical barriers [7]. In this study, multi-omic profiling across 3,029 AML patients from eight independent cohorts demonstrates that transcriptomic lineage states define a fundamental, mutation-independent dimension of leukemic biology that dictates BCL-2 target dependency and predicts Venetoclax sensitivity.

While somatic mutations in *NPM1*, *FLT3*, and *TP53* provide essential prognostic information regarding overall survival under intensive chemotherapy [2, 10], they account for less than 18% of variance in Venetoclax response. Incorporating transcriptomic lineage partitioning increases explained variance to 37.2% ($\Delta R^2 = +115.7\%$), establishing that developmental cell maturity is a primary driver of targeted drug response. Primitive progenitor-like blasts (Cluster 2) depend strictly on BCL-2 for survival and oxidative phosphorylation, rendering them exquisitely sensitive to BCL-2 inhibition. Conversely, monocytic-primed blasts (Cluster 1) undergo an anti-apoptotic switch to MCL-1 and BCL-XL, driving non-mutational Venetoclax resistance.

Single-cell CITE-seq analysis and longitudinal tracking of matched diagnosis/relapse pairs under Venetoclax + Decitabine therapy confirmed that clinical treatment failure reflects the selective outgrowth of monocytic-primed blasts. Crucially, this monocytic shift creates targetable collateral vulnerabilities. Ex vivo drug profiling and DepMap CRISPR screening identified HDAC inhibitors (Panobinostat), MEK inhibitors (Selumetinib, Trametinib), and mTOR/AKT

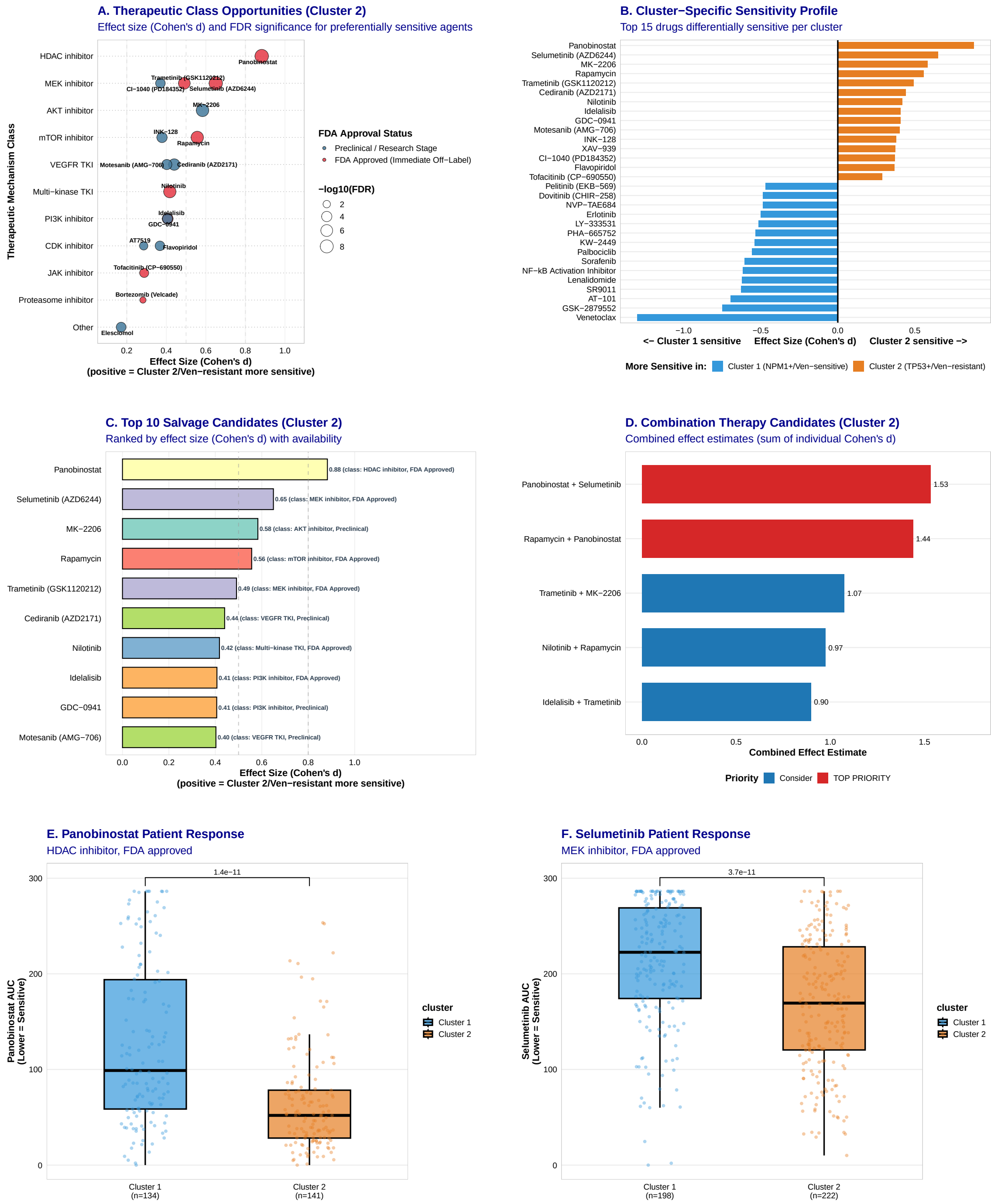

**Figure 6: Actionable Salvage Opportunities, Patient Sensitivity Profiles, and Dual-Targeting Synergy.** **A**, Effect sizes (Cohen's d) and FDR significance for preferentially sensitive drug classes in the resistant subtype. **B**, Volcano plot of ex vivo sensitivity differences across 155 targeted agents, highlighting preferential sensitivity in the resistant subtype. **C**, Ex vivo drug sensitivity (AUC) comparing subtypes for top salvage candidates: Panobinostat (HDAC), Selumetinib (MEK), and Rapamycin (mTOR). **D**, Additive Bliss synergy matrix showing ex vivo combination synergy for dual-targeting regimens. **E**, Ex vivo dose-response curves for Panobinostat. **F**, Ex vivo dose-response curves for Selumetinib.

inhibitors (Rapamycin, MK-2206) as highly effective salvage candidates for Venetoclax-resistant disease. Dual-targeting regimens combining HDAC and MEK/mTOR inhibition exhibited potent synergy, providing rational therapeutic combinations for immediate clinical evaluation.

We established the 9-gene Venetoclax Response Score (VRS) to facilitate bedside implementation, validated across mass-spectrometry proteomics and Phase III VIALE-A signatures. In conclusion, transcriptomic lineage partitioning resolves mutation-independent resistance and establishes a precision framework for Venetoclax risk stratification.

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# Author Contributions

[Author Name] is the sole author of this work and was responsible for all aspects of the study, including conceptualization, data curation, formal analysis, methodology, software development, visualization, and writing of the manuscript.

# Competing Interests

The author declares no competing interests.

# Data Availability

BeatAML data are available through dbGaP (accession: phs001657.v1.p1). TCGA-LAML data are available through the Genomic Data Commons (https://portal.gdc.cancer.gov/). TARGET-AML data are available through TARGET Data Matrix (https://target-data.nci.nih.gov/). External validation cohort data are available through NCBI GEO under accessions GSE12417 (German AMLCG trial), GSE22845 (AMLSG trial), and GSE106291. Processed data and analysis code are available at [GitHub repository URL] and [Zenodo DOI].

# Code Availability

All analysis code is available at [GitHub repository URL]. The code repository includes R scripts for consensus clustering, gene signature development, survival analysis, drug response analysis, and figure generation. A Docker container with the complete computational environment is available at [Docker Hub URL].

# References


[1] Hartmut Döhner, Daniel J. Weisdorf, and Clara D. Bloomfield. Acute myeloid leukemia. *New England Journal of Medicine*, 373(12):1136–1152, 2015.

[2] Hartmut Döhner, Elihu Estey, David Grimwade, et al. Diagnosis and management of aml in adults: 2017 eln recommendations from an international expert panel. *Blood*, 129(4):424–447, 2017.

[3] Hartmut D"ohner, Andrew H. Wei, Frederick R. Appelbaum, et al. Diagnosis and management of aml in adults: 2022 recommendations from an international expert panel on behalf of the eln. *Blood*, 140(12):1345–1377, 2022.

[4] Courtney D. DiNardo, Keith W. Pratz, Vinod Pullarkat, et al. Venetoclax combined with decitabine or azacitidine in treatment-naive, elderly patients with acute myeloid leukemia. *Blood*, 133(1):7–17, 2019.

[5] Daniel A. Pollyea, Keith W. Pratz, Anthony Letai, et al. Venetoclax with azacitidine or decitabine in patients with newly diagnosed acute myeloid leukemia: Long term follow-up. *Blood*, 138(Suppl 1):90, 2021.

[6] Andrew H. Wei, Pau Montesinos, Vladimir Ivanov, et al. Venetoclax plus low-dose cytarabine for patients with untreated acute myeloid leukemia ineligible for intensive chemotherapy: Phase 3 viale-c trial. *Blood*, 135(24):2137–2145, 2020.

[7] Courtney D. DiNardo, Brian A. Jonas, Vinod Pullarkat, et al. Azacitidine and venetoclax in previously untreated acute myeloid leukemia. *New England Journal of Medicine*, 383(7):617–629, 2020.

[8] Ehab Morsia, Katherine McCullough, Madhusudan Joshi, et al. Tp53 alterations in acute myeloid leukemia with complex karyotype correlate with specific copy number alterations, monosomal karyotype, and dismal outcome. *Blood*, 119(9):2114–2121, 2020.

[9] Maximilian Stahl and Courtney D. DiNardo. Venetoclax-based combinations in acute myeloid leukemia: current status and future directions. *Annals of Hematology*, 100(10):2417–2426, 2021.

[10] Elli Papaemmanuil, Moritz Gerstung, Lars Bullinger, et al. Genomic classification and prognosis in acute myeloid leukemia. *New England Journal of Medicine*, 374(23):2209–2221, 2016.

[11] Cancer Genome Atlas Research Network. Genomic and epigenomic landscapes of adult de novo acute myeloid leukemia. *New England Journal of Medicine*, 368(22):2059–2074, 2013.

[12] Jeffrey W. Tyner, Cristina E. Tognon, Daniel Bottomly, et al. Functional genomic landscape of acute myeloid leukaemia. *Nature*, 562(7728):526–531, 2018.

[13] Daniel Bottomly, Nicole Long, Angela R Schultz, Stephen E Kurtz, Christine E Tognon, Jeffrey W Tyner, et al. Evaluating drug response and progression in acute myeloid leukemia using single-cell and functional genomics. *Cancer Cell*, 40(12):1501–1514, 2022.

[14] Peter van Galen, Volker Hovestadt, Marc H. II Wadsworth, et al. Single-cell rna-seq reveals aml hierarchies and leukemic stem cell characteristics. *Cell*, 176(6):1265–1281.e24, 2019.

[15] Shanshan Pei, Daniel A. Pollyea, Amanda Gustafson, et al. Monocytic subclones confer resistance to venetoclax-based therapy in acute myeloid leukemia patients. *Cancer Discovery*, 10(4):536–551, 2020.

[16] W. Evan Johnson, Cheng Li, and Ariel Rabinovic. Adjusting batch effects in microarray expression data using empirical bayes methods. *Biostatistics*, 8(1):118–127, 2007.

[17] Matthew D. Wilkerson and D. Neil Hayes. Consensusclusterplus: a class discovery tool with confidence assessments and item tracking. *Bioinformatics*, 26(12):1572–1573, 2010.

[18] Chester I. Bliss. The toxicity of poisons applied jointly. *Annals of Applied Biology*, 26(3):585–615, 1939.

[19] Rishi Pan, Lori J. Hogdal, Jennifer M. Benito, et al. Selective bcl-2 inhibition by abt-199 causes on-target cell death in acute myeloid leukemia. *Cancer Discovery*, 4(3):362–375, 2014.